

Problem 1

a) Yes, it's a binomial distribution setting.

Reasons

- Each question is answered correctly or not.
- All 10 questions are unrelated, so independent.
- The probability of answering correctly a question is $p = \frac{1}{5}$, for each of the 10 questions.

b) Let $X = \#$ questions answered correctly.

Then, $X \sim \text{Bin}(n=10, p=\frac{1}{5}=0.2)$

$$\begin{aligned} IP(X=5) &= \binom{10}{5} (0.2)^5 (1-0.2)^5 \\ &\approx 0.0264 \end{aligned}$$

$$\text{Note that } \binom{10}{5} = \frac{10!}{5!(10-5)!} = 252.$$

Problem 2

Let $X = \#$ of bulbs that fail within 50,000 hours out of 10 bulbs. Then,

$$X \sim \text{Bin}(n=10, p=0.25)$$

$$a) \quad IP(X < 2) = IP(X \leq 1)$$

$$\begin{aligned} &= \binom{10}{1} 0.25^1 (1-0.25)^{10-1} \\ &\approx 0.2440. \end{aligned}$$

$$b) \quad IP(X > 4) = 1 - IP(X \leq 4)$$

$$= 1 - \sum_{k=0}^4 \binom{10}{k} 0.25^k (1-0.25)^{10-k}$$

$$\begin{aligned} &= 1 - \left\{ \binom{10}{0} 0.25^0 (0.75)^{10} + \right. \\ &\quad \binom{10}{1} 0.25^1 (0.75)^{10-1} + \\ &\quad \binom{10}{2} 0.25^2 (0.75)^{10-2} + \\ &\quad \binom{10}{3} 0.25^3 (0.75)^{10-3} + \\ &\quad \left. \binom{10}{4} 0.25^4 (0.75)^{10-4} \right\} \end{aligned}$$

$$\approx 0.0781.$$

c) Let find the expected value:

$$E[X] = np = 10 \times 0.25 = 2.5$$

Hence, on average, we expect roughly 3 light bulbs to fail within 50,000 hours.

Problem 3

Let $X = \#$ calls received per day. Then,

$$X \sim \text{Poi}(\lambda = 6)$$

$$\begin{aligned} \text{a) } IP(X \leq 2) &= \sum_{k=0}^2 \frac{e^{-6} \cdot 6^k}{k!} \\ &= e^{-6} \left\{ \frac{6^0}{0!} + \frac{6^1}{1!} + \frac{6^2}{2!} \right\} \end{aligned}$$

$$\approx 0.0620.$$

$$\begin{aligned} \text{b) } IP(X(2) = 5) &= \frac{e^{-2 \times 6} \cdot (2 \times 6)^5}{5!} \\ &= \frac{e^{-12} \cdot 12^5}{5!} \end{aligned}$$

$$\approx 0.0127.$$

Note that $X(2) \sim \text{Poi}(\lambda = 2 \times 6 = 12)$.

Problem 4

Let $Y = \#$ of cars per year that will experience the catastrophe. Then,

$$Y \sim \text{Poi}(\lambda = 2.2).$$

$$\begin{aligned} \text{a) } \text{IP}(Y=0) &= \frac{e^{-2.2} \times 2.2^0}{0!} \\ &\approx 0.1108 \end{aligned}$$

$$\text{b) } \text{IP}(Y \geq 3) = 1 - \text{IP}(X < 3)$$

$$= 1 - \text{IP}(X \leq 2)$$

$$= 1 - \sum_{k=0}^2 \frac{e^{-2.2} \cdot (2.2)^k}{k!}$$

$$= 1 - e^{-2.2} \cdot \left\{ \frac{(2.2)^0}{0!} + \frac{(2.2)^1}{1!} + \frac{(2.2)^2}{2!} \right\}$$

$$\approx 0.3773.$$

Problem 5

X discrete R.V with pmf given by:

$$f(x) = \begin{cases} \frac{1}{12} (6 - 2x), & x = 0, 1, 2 \\ 0 & \text{otherwise.} \end{cases}$$

a) Let's show that $f(x)$ is a valid pmf.

The pmf table is given by:

x	0	1	2
$f(x)$	$6/12$ " $1/2$	$4/12$ " $1/3$	$2/12$ " $1/6$

$$\bullet \forall x \in \{0, 1, 2\}, f(x) \geq 0$$

$$\bullet \sum_{x=0}^2 f(x) = \frac{6}{12} + \frac{4}{12} + \frac{2}{12} = \frac{12}{12} = 1.$$

b) Let's compute $E[X]$.

$$E[X] = \sum_{x=0}^2 x f(x)$$

$$= 0 \times \left(\frac{1}{2}\right) + 1 \left(\frac{1}{3}\right) + 2 \left(\frac{1}{6}\right) = \frac{2}{3}.$$

Problem 6

Let $X = \#$ animals per hour. Then,
 $X \sim \text{Poi}(\lambda = 11)$.

$$\begin{aligned} \text{a) } IP(X > 3) &= 1 - IP(X \leq 3) \\ &= 1 - \sum_{k=0}^3 \frac{e^{-11} \cdot 11^k}{k!} \\ &= 1 - e^{-11} \cdot \left\{ \frac{11^0}{0!} + \frac{11^1}{1!} + \frac{11^2}{2!} + \frac{11^3}{3!} \right\} \\ &= 0.9951 \end{aligned}$$

$$\text{b) } IP(X = 5) = \frac{e^{-11} \cdot 11^5}{5!} \approx 0.0224$$

Problem 7

X is a R.V with pdf $f(x) = \begin{cases} \frac{1}{4.5} e^{-x/4.5}, & x \geq 0 \\ 0 & , x < 0. \end{cases}$

a) Let's find the CDF of X .

- Observe that $X \sim \text{Exp}(\lambda = 1/4.5)$

$$\Rightarrow F(x) = \begin{cases} 1 - e^{-x/4.5}, & \forall x \geq 0 \\ 0 & , \forall x < 0 \end{cases}$$

- For $x < 0$,

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(u) du \\ &= \int_{-\infty}^x 0 du = 0. \end{aligned}$$

- For $x \geq 0$,

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(u) du \\ &= \int_{-\infty}^0 f(u) du + \int_0^x f(u) du \end{aligned}$$

$$\begin{aligned}
 &= 0 + \int_0^x \frac{1}{4.5} e^{-u/4.5} du \\
 &= \left[-e^{-u/4.5} \right]_0^x \\
 &= 1 - e^{-x/4.5}
 \end{aligned}$$

Аналогично, $F(x) = \begin{cases} 1 - e^{-x/4.5}, & \forall x \geq 0 \\ 0, & \forall x < 0. \end{cases}$

$$\begin{aligned}
 \text{b) } P(X \leq 2.5) &= F(2.5) \\
 &= 1 - e^{-2.5/4.5} \\
 &\approx 0.4262.
 \end{aligned}$$

$$\begin{aligned}
 \text{c) } P(X \geq 6.75) &= e^{-6.75/4.5} \\
 &\approx 0.2231
 \end{aligned}$$

$$\text{d) } E[X] = \frac{1}{\frac{1}{4.5}} = 4.5.$$

Problem 8

Let's find the value of c that makes the following valid pdf:

a) $f(x) = cx^2, 0 < x < 4.$

$$\begin{aligned}\int_0^4 f(x) dx &= \int_0^4 cx^2 dx = c \int_0^4 x^2 dx \\ &= c \left[\frac{1}{3} x^3 \right]_0^4 = \frac{c4^3}{3} = 1\end{aligned}$$

$$\Rightarrow c = \frac{3}{4^3} = \frac{3}{64}.$$

b) $f(x) = c(1+2x), 0 < x < 2$

$$\int_0^2 f(x) dx = \int_0^2 c(1+2x) dx$$

$$= c \int_0^2 (1+2x) dx$$

$$= c \left[x + x^2 \right]_0^2$$

$$= c(2+2^2) = c(6) = 6c = 1$$

$$\Rightarrow c = \frac{1}{6}$$

$$c) f(x) = c e^{-x}, x > 0$$

$$\int_0^{\infty} c e^{-x} dx = c \int_0^{\infty} e^{-x} dx$$

$$= c [-e^{-x}]_0^{\infty}$$

$$= c(1) = 1 \Rightarrow c = \boxed{1}$$

Problem 9

False

Problem 10

e)

Problem 11

True

Problem 12

a) Let's find $F(x)$.

$$F(x) = IP(X \leq x), \quad \forall x \in \mathbb{R}.$$

$$= \sum_{\text{all } y \leq x} f(y)$$

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.1 & 0 \leq x < 1 \\ 0.25 & 1 \leq x < 2 \\ 0.50 & 2 \leq x < 3 \\ 0.75 & 3 \leq x < 4 \\ 0.90 & 4 \leq x < 5 \\ 1 & x \geq 5 \end{cases}$$

$$b) E[X] = \sum_{\text{all } x} x f(x)$$

$$= 0(0.1) + 1(0.15) + 2(0.25)$$

$$+ 3(0.15) + 4(0.15) + 5(0.1)$$

$$= 2.5$$

$$c) IP(X \leq 3.5) = F(3.5) = 0.75$$

Problem 13

$$\begin{aligned} E[X^2] &= \int_3^{\infty} x^2 f(x) dx \\ &= \int_3^{\infty} x^2 \cdot (5e^{-5(x-3)}) dx \\ &= \int_3^{\infty} 5x^2 e^{-5(x-3)} dx \\ &= b) \end{aligned}$$

Problem 14

b) About 95% of scores are between 72 and 128.

Problem 15

False

Problem 16

$$E[X] = \int_1^{\infty} x f(x) dx = \int_1^{\infty} x \cdot \frac{2}{x^3} dx = \int_1^{\infty} \frac{2}{x^2} dx = b)$$

Problem 17

Let $X \sim \text{Bin}(n=7, p=0.6)$.

$$\begin{aligned} IP(3 \leq X \leq 5) &= \sum_{x=3}^5 p(x) \\ &= p(3) + p(4) + p(5) \\ &= \binom{7}{3} 0.6^3 (0.3)^{7-3} + \binom{7}{4} 0.6^4 (0.3)^{7-4} \\ &\quad + \binom{7}{5} 0.6^5 (0.3)^{7-5} \\ &= 0.7451 \end{aligned}$$

Problem 18

X = # of drivers who travel between a particular origin and destination during a designated time window. Then:

$$X \sim \text{Poi}(\lambda = 20).$$

$$\begin{aligned} \text{a) } IP(X \leq 10) &= F(10) \\ &= \sum_{k=1}^{10} \frac{e^{-20} \cdot 20^k}{k!} \end{aligned}$$

$$\approx 0.0108$$

$$\begin{aligned} b) \quad IP(X > 20) &= 1 - IP(X \leq 20) \\ &= 1 - \sum_{k=0}^{20} \frac{e^{-20} \cdot 20^k}{k!} \end{aligned}$$

$$\approx 0.4409.$$

$$c) \quad IP(10 < X < 20) = \sum_{k=11}^{19} \frac{e^{-20} \cdot 20^k}{k!}$$

$$\approx 0.4594.$$

Problem 19

μ	σ	$\mu + \sigma$	$\mu + 2\sigma$	$\mu + 3\sigma$
48	2	50	52	54

Problem 20

c) 50% (because symmetric)

Problem 21

d) 84%

Problem 22

$$z = \frac{25 - 30}{5} = -1 = b)$$

Problem 23

$$x = 5 \times (-2) + 30 = 20 = a)$$

Problem 24

True because negative and positive numbers are equally covered, that's 50%-50%. Hence, a negative number would have its right-hand area greater than 50%.

Problem 25

True because symmetric distribution.

Problem 26

a) per the Central Limit Theorem. ($n \geq 30$)

Problem 27

b) False since $n < 30$ and population distribution not normally distributed.

Problem 28

True since $\int_1^3 f(x) dx = 1$
= Area under the curve.

Problem 29

Discrete Random Variable.

Problem 30

$$\sum_{\text{all } x} p(x) = 1 \Rightarrow k + 0.15 + 0.15 + 5k + 0.1 = 1$$

$$\Rightarrow 6k + 0.4 = 1$$

$$\Rightarrow k = \frac{1 - 0.4}{6} = 0.1.$$

Problem 31

Let's find:

$$\begin{aligned} \bullet \text{ } IP(X \leq 3.5) &= F(3.5) \\ &= 0.53 \end{aligned}$$

$$\begin{aligned} \bullet \text{ } IP(3 \leq X \leq 6) &= IP(X \leq 6) - IP(X \leq 2) \\ &= F(6) - F(2) \\ &= 0.87 - 0.40 \\ &= 0.47 \end{aligned}$$

$$\begin{aligned} \bullet \text{ } IP(X \geq 4) &= 1 - IP(X < 4) \\ &= 1 - IP(X \leq 3) \\ &= 1 - F(3) \\ &= 1 - 0.53 \\ &= 0.47. \end{aligned}$$

The End!